

DESIGN AND ANALYSIS OF ALGORITHMS

Question Paper Code: 526430

Course Code: CS23431

Semester: IV Semester

Regulation: 2023

Time: 3 Hours

Maximum Marks: 100

PART A — (10 × 2 = 20 Marks)

1. Define an algorithm. What are the key characteristics of a good algorithm?
2. State any two properties of Big-O notation.
3. Outline the advantages of Binary Search.
4. List a few limitations of Heap Sort.
5. State the principle of optimality in Dynamic Programming.
6. What is a greedy algorithm? Mention one limitation.
7. Define Maximum Flow Problem.
8. Define Stable Marriage Problem.
9. Illustrate NP-Complete Problem.
10. Define Backtracking. Give one example problem.

PART B — (5 × 13 = 65 Marks)

11(a). Explain the fundamentals of algorithmic problem solving with a suitable example.

OR

11(b). Analyze the time complexity of recursive and non-recursive algorithms with examples.

12(a). Apply the brute force approach for solving the string matching problem with an example.

OR

12(b). Describe the Divide and Conquer approach and also explain Merge Sort with analysis.

13(a). Solve the Coin Changing Problem using Dynamic Programming and explain the steps.

OR

13(b). Explain Kruskal's Algorithm and analyze its time complexity.

14(a). Explain the Simplex Method for solving Linear Programming Problems.

OR

14(b). Describe the Maximum Matching Problem in Bipartite Graphs with an example.

15(a). Explain in detail the LIFO-search and FIFO-search.

OR

15(b). Solve the N-Queen Problem using Backtracking with explanation.

PART C — (1 × 15 = 15 Marks)

16(a). Apply Branch and Bound technique and solve the Travelling Salesman Problem using it.

OR

16(b). Evaluate Approximation Algorithms for NP-Hard Problems and explain with Knapsack Problem.